1.KEYBOARD



- An alphanumeric keyboard on a graphics system is used primarily as a **device for entering text.**
- The keyboard is an efficient device for inputting such nongraphic data such as **picture labels** associated with a graphics display.
- Keyboards can also be provided with features to enter **screen coordinates, menu selection,** etc.
- **Keyboards** are also **used for computer gaming.**

Other types of cursor-positioning devices, such as a trackball or joystick, are included

on some keyboards.

Additionally, a numeric keypad is, often included on the key- board for fast entry of numeric data.

2. MOUSE

- A mouse is small **hand-held DEVICE** used to position the screen cursor.
- Wheels or rollers on the bottom of the mouse can be used to record the amount and direction of movement OF THE CURSOR.
- The optical sensor detects movement

  Across the lines in the grid.



- Physically, a mouse consists of an object held in one's hand, with one or more buttons. Mice often also feature other elements, such as touch surfaces and "wheels", which enable additional control and dimensional input.

3. Trackball and Spaceball

- A **trackball** is a ball that can be rotated with the fingers or palm of the hand to produce cursor movement.
- Potentiometers, attached to the trackball, measure the amount and direction of rotation for movement.



spaceball



- While a trackball is a 2D positioning device, a spaceball provides 3D positioning.
- A spaceball does not actually move. Strain gauges measure the amount of pressure applied to the spaceball to provide positioning details.
- Spaceballs are used for three-dimensional positioning and selection operations in virtual-reality systems and animation systems etc.

4. joysticks

- A **joystick** is an input device consisting of a stick that pivots on a base and reports its angle or direction to the device it is controlling.
- Joysticks are often used to control video games, and usually have one or more push-buttons whose state can also be read by the computer.



- Potentiometers mounted at the base of the joystick measure the amount of movement, and springs return the stick to the center position when it is released.
- Joysticks are also used for controlling machines such as cranes, trucks, etc.

5. Image Scanners

- Image scanner is a device that optically scans images, printed text, handwriting or an object and converts it to a digital image.
- Once we have the internal representation of a picture, we can apply transformations to rotate, scale, or crop the picture to a particular screen area.



6. Touch Panels

- Touch panels allow displayed objects or screen positions to be selected with the touch of a finger.
- A user can give input or control the information processing system through simple or multi-touch gestures by touching the screen with a special stylus and/or one or more fingers.



- The touchscreen enables the user to interact directly with what is displayed, rather than using a mouse, touchpad, or any other such device.
- The opposite vertical and horizontal edges contain light detectors.
- These detectors are used to record which beams are interrupted when the panel is touched.

7. Light Pens



- Light pens are pencil-shaped devices are used to select screen positions by detecting the light coming from points on the CRT screen.
- It allows the user to point to displayed objects or draw on the screen in a similar way to a touchscreen but with greater positional accuracy.
- A light pen detects a change of brightness of nearby screen pixels when scanned by cathode ray tube electron beam and communicates the timing of this event to the computer

- The first light pen was created around 1955 as part of the Whirlwind project at MIT.
- Because the user was required to hold their arm in front of the screen for long periods of time (potentially causing "gorilla arm") or to use a desk that tilts the monitor, the light pen fell out of use as a general purpose input device.

DIGITIZER



DESCRIPTION OF DIGITIZER

- Digitizer is an input device, which converts analog information into a digital form.
- Digitizer can convert a signal from the television camera into a series of numbers that could be stored in a computer. They can be used by the computer to create a picture of whatever the camera had been pointed at.
- Digitizer is also known as Tablet or Graphics Tablet because it converts graphics and pictorial data into binary inputs. A graphic tablet as digitizer is used for doing fine works of drawing and images manipulation applications.

MAGNETIC INK CARD READER (MICR)



MAGNETIC INK CARD READER (MICR)

- MICR input device is generally used in banks because of a large number of cheques to be processed everyday. The bank's code number and cheque number are printed on the cheques with a special type of ink that contains particles of magnetic material that are machine readable.

- This reading process is called Magnetic Ink Character Recognition (MICR). The main advantage of MICR is that it is fast and less error prone.